Results and Documentation Guide

What to record, how to name it, and how to package it for the paper

1. Evidence folder structure

```
evidence/
  hardware/     photos, wiring, serial logs, camera_tests.csv, pan_tilt_tests.csv
  data/         DATA_SOURCES.txt, dataset_log.csv, class_counts.csv, split_counts.csv
  detector/     runs/plant_v1/ (all outputs), detector_results.csv, acceptance_test.csv
  classifier/   classifier_seed0..2.json, confusion PNGs, classifier_results.csv
  crossdomain/  plantdoc_*.json, own_esp32cam_*.json, class_mapping.csv
  speed/        latency output text, link_latency_log.csv
  figures/      final PNGs named fig_<topic>_vN.png
  notes/        daily_log.md (date, what you did, what failed, what you changed)
```

2. Naming convention

`YYYYMMDD_topic_detail_vN`, for example `20261108_classifier_plantdoc_seed1_v1.json`. Never overwrite a result; make a new version. Keep a daily log (even 5 lines) in notes/daily_log.md.

3. Tables to fill (CSV templates are in the Templates folder)

Template	What goes in it	Becomes in the paper
camera_tests.csv	Resolution, quality, distance, lighting, frame time, file size	Camera subsection / table
pan_tilt_tests.csv	Command vs measured angle, jitter, brownouts	Hardware validation text
dataset_log.csv	One row per image	Dataset description, counts
detector_results.csv	P, R, mAP50, mAP50-95, latency	Table II
classifier_results.csv	Accuracy, macro-F1 per test set and seed	Table III
link_latency_log.csv	RTT, loss, RSSI, snapshot delivery over WiFi / 4G	Table IV (with Shreyas)
literature_table.csv	15-20 verified references	Related work

4. Figure list (make these as clean PNGs, 300 dpi)

Fig	Content	Tips
F1	Photo of the pan-tilt camera mounted on the rover	No faces/names; good lighting
F2	Wiring diagram of camera node (hand-drawn is fine, or draw.io)	Label pins GPIO 13/14, separate servo supply
F3	Example detections (4-6 frames with boxes and confidence)	Include one failure
F4	Detector PR curve and training curves	From runs/plant_v1
F5	Classifier confusion matrix (in-domain) and cross-domain	Same class order; readable fonts
F6	Accuracy bar chart: in-domain vs cross-domain vs ESP32-CAM	Error bars from 3 seeds
F7	Grid of correct vs wrong classifications	Shows honest failure modes
F8	Sample alert record (what the farmer sees on the dashboard)	Screenshot from Shreyas's dashboard

Chart rules: label axes and units, use colour-blind friendly colours, font size at least 8 pt after scaling to one column (about 8.8 cm wide), write the caption yourself in one factual sentence.

5. The numbers Shreyas needs, in one table

Quantity	Your value (fill in)	Source file
Public tomato images used (train / val / test)		split_counts.csv
Own images (healthy / each disease), number of plants		class_counts.csv

Quantity	Your value (fill in)	Source file
Detector precision / recall / mAP50 / mAP50-95		detector_results.csv
Classifier accuracy, in-domain (mean +/- std)		classifier_results.csv
Classifier accuracy, cross-domain PlantDoc		classifier_results.csv
Classifier accuracy, own ESP32-CAM set		classifier_results.csv
Macro-F1 for each of the three tests		classifier_results.csv
Camera snapshot time (median / p95, ms)		latency output
Classifier inference time (median / p95, ms) and device		latency output
Detector inference time (median, ms) and device		latency output

RULE: Write the one-page 'results summary' last, from the CSVs. If a number in the summary does not match a CSV, fix the summary.